

- 1 (a) (i) Take direction up the slope as positive.

$$v^2 = u^2 + 2as_0$$

$$0 = 7.0^2 + 2(-9.81 \sin 30^\circ)s_0$$

$$s_0 = 4.9949 = 4.99 \text{ m}$$

OR

By the principle of conservation of energy,
increase in G.P.E. = decrease in K.E.

$$mg(\Delta h) = \frac{1}{2}mv^2 - 0, \text{ where } \Delta h \text{ is the max. vertical height from ground}$$

$$\Delta h = \frac{v^2}{2g}$$

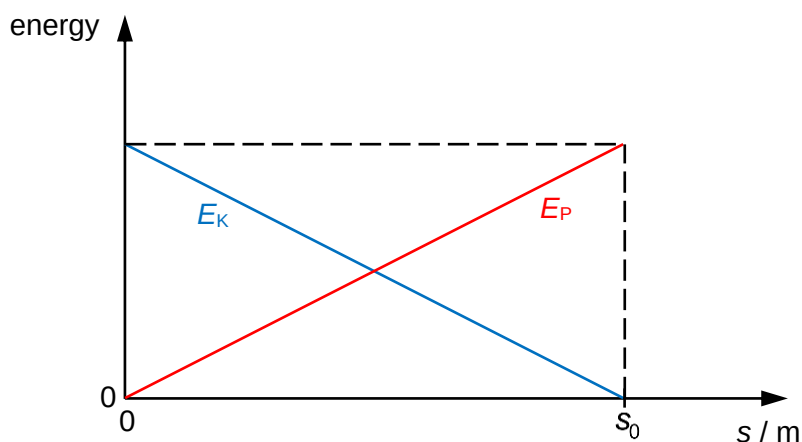
$$= \frac{7.0^2}{2(9.81)}$$

$$s_0 = \frac{\Delta h}{\sin 30^\circ}$$

$$= \frac{7.0^2}{2(9.81)} \div \sin 30^\circ$$

$$= 4.9949 = 4.99 \text{ m}$$

(ii)



*Both graphs aligned at the same total energy and s_0 and clearly labelled.

$$1. \quad E_K = \frac{1}{2}mv^2 = \frac{1}{2}m(u^2 + 2[-g \sin \theta]s)$$

$$E_K = \frac{1}{2}mu^2 - (mg \sin \theta)s$$

Graph of E_K - s is a straight line with negative gradient and vertical intercept

$$\frac{1}{2}mu^2.$$

$$2. \quad E_P = mg(\Delta h) = mg(s \sin \theta)$$

$$E_P = (mg \sin \theta)s$$

Graph of E_P - s is a straight line through the origin with positive gradient.

Raffles Institution
Year 5-6 Physics Department

(b) (i) $v^2 = u^2 + 2as$

$$v^2 = 14.0^2 + 2(-9.81 \sin 30^\circ) \left(\frac{4.0}{\sin 30^\circ} \right)$$

$$v = 10.841 = 10.8 \text{ m s}^{-1} \quad (\text{shown})$$

OR

decrease in K.E. = increase in G.P.E.

$$\frac{1}{2}mu^2 - \frac{1}{2}mv^2 = mg(\Delta h)$$

$$v^2 = u^2 - 2g(\Delta h) = 14.0^2 - 2(9.81)(4.0)$$

$$v = 10.841 = 10.8 \text{ m s}^{-1} \quad (\text{shown})$$

(ii) Take directions to the right and upwards as positive.

Time of flight after the ball leaves the top of the slope to the ground:

$$s_y = u_y t + \frac{1}{2}a_y t^2$$

$$-4.0 = (10.8 \sin 30^\circ)t + \frac{1}{2}(-9.81)t^2$$

$$t = 1.6080 \text{ s or } -0.50713 \text{ s (NA)}$$

Horizontal distance travelled:

$$s_x = u_x t + \frac{1}{2}a_x t^2$$

$$= (10.8 \cos 30^\circ)(1.6080) + 0$$

$$= 15.040 = 15.0 \text{ m}$$

*State both values of t and reject the negative one.

Raffles Institution
Year 5-6 Physics Department

- 2** **(a)** The initial total momentum of both balls is not zero